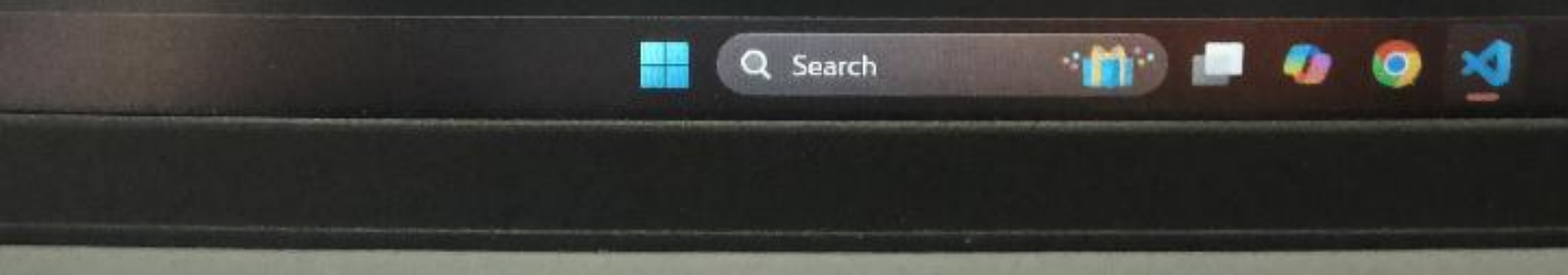
Week2 > LinearSearch.java > LinearSearch > main(String[])

```java
package Week2;
import java.util.*;

public class LinearSearch{
    Run | Debug
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int N = sc.nextInt();
        int[] transactions = new int[N];

        for(int i=0;i<N;i++){
            transactions[i] = sc.nextInt();
        }

        int suspiciousId = sc.nextInt();

        boolean found = false;

        for(int i=0;i<N;i++){
            if(transactions[i] ==  suspiciousId){
                found = true;
                break;
            }
        }

        if(found){
            System.out.println(x: "Fraud Transaction found");
        }else{
                System.out.println(x: "Fraud Transaction NOT found");
        }

        sc.close();
    }
}
```



```java
package Week2;
import java.util.*;

public class BinarySearch {
    Run | Debug
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int N = sc.nextInt();
        int[] logID = new int[N];

        for(int i=0;i<N;i++){
            logID[i] = sc.nextInt();
        }

        int targetID = sc.nextInt();
        int low = 0;
        int high =logID.length-1;
        boolean found = false;

        while(low<high){
            int mid = low+(high-low)/2;
            if(logID[mid]==targetID){
                found= true;
                break;
            }else if(logID[mid]<targetID){
                low=mid+1;
            }else{
                high=mid-1;
            }
        }

        if(found){
            System.out.println(x: "log found");
        }else{
            System.out.println(x: "log  not found");
        }

        sc.close();
    }
}
```

LinearSearch.java | BubbleSort.java | LinearSearch.class

Week2 > BubbleSort.java > BubbleSort > main(String[])

```java
package Week2;
import java.util.*;

public class BubbleSort {
    Run | Debug
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.println(x: "Enter number of salary:");
        int N = sc.nextInt();

        int[] salary = new int[N];
        System.out.println(x: "Enter salary:");
        for(int i=0;i<N;i++){
            salary[i] = sc.nextInt();
        }
        // sorting using bubble sort
        boolean swapped = false;
        for(int i=0;i<N;i++){
         for(int j=0;j<N-1-i;j++){
            if(salary[j]>salary[j+1]){
                int temp = salary[j+1];
                salary[j+1]=salary[j];
                salary[j]=temp;
                swapped=true;
            }
         }
         if(!swapped){
            break;
         }
        }
        //printing of sorted array
        System.out.println(x: "sorted salary: ");
        for(int i=0;i<N;i++){
           System.out.println(salary[i]);
        }
        sc.close();
    }
}
```

Week2 > SelectionSort.java > SelectionSort > main(String[])

```java
package Week2;
import java.util.*;

public class SelectionSort {
    Run | Debug
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.println(x: "Enter number of priority:");
        int N = sc.nextInt();

        int[] priority = new int[N];
        System.out.println(x: "Enter priority:");
        for(int i=0;i<N;i++){
            priority[i] = sc.nextInt();
        }

        // sorting using selection
        for(int i=0;i<priority.length-1;i++){
            int minPos = i;
            for(int j=i+1;j<priority.length;j++){
                if(priority[minPos]>priority[j]){
                    minPos=j;
                }
            }
            int temp =priority[minPos];
            priority[minPos]=priority[i];
            priority[i]=temp;
        }

        //printing of sorted array
        System.out.println(x: "sorted priority: ");
        for(int i=0;i<N;i++){
            System.out.println(priority[i]);
        }
        sc.close();

    }
}
```

LinearSearch.java | SelectionSort.java | InsertionSort.java | LinearSearch.class

Week2 > InsertionSort.java > ...

```java
package Week2;
import java.util.*;

public class InsertionSort {
    Run | Debug
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.println(x: "Enter number of response time:");
        int N = sc.nextInt();

        int[] response = new int[N];
        System.out.println(x: "Enter response time:");
        for(int i=0;i<N;i++){
            response[i] = sc.nextInt();
        }

        // sorting using innsertion sort
        for(int i=1;i<response.length;i++){
            int curr = response[i];
            int prev= i-1;
            while(prev>=0 && response[prev]>curr){
                response[prev+1]=response[prev];
                prev--;
            }
            response[prev+1]=curr;
        }


        //printing of sorted array
        System.out.println(x: "sorted response time: ");
        for(int i=0;i<N;i++){
            System.out.println(response[i]);
        }
        sc.close();

    }

}
```

Integer.MAX_VALUE (1-)

```
update(marks, a);
System.out.println(a);
for (int i=0; i<marks.length; i++){
    System.out.print(mark[i]+" ");
}
}
```

Linear Search

```
public static int linearSearch(int numbers[], int Key){
    for(int i=0; i<numbers.length; i++){
        if(numbers[i]==Key){
            return i;
        }
    }
    return -1;
}
```

```
public static void main(String args[]){
    int numbers[]={2,4,6,8,10,12,14,16};
    int Key=10;
    int index = linearSearch(numbers, Key);
    if(index==-1){
        Sys("NOT found");
    }else{
        Sys("Key is at index: "+index);
    }
}
```

Binary Search

```
public static int binarySearch(int numbers[], int key){
    int start = 0, end = numbers.length - 1;
    while(start <= end){
        int mid = (start + end) / 2;
        if (numbers[mid] == key){
            return mid;
        }
        if (numbers[mid] < key){
            start = mid + 1;
        } else{
            end = mid - 1;
        }
    }
    return -1;
}
public static void main(String args[]){
    int numbers[] = {2, 4, 6, 8, 10, 12, 14};
    int key = 4;
    System.out.println("index for key is: " + binarySearch(numbers, key));
}
```

Bubble Sort

```
public static void bubbleSort(int arr[]){
    int swapped=0; for(int i=0; i<arr.length-1; i++){
                   for(int j=0; j<arr.length-i-1; j++){
                       if(arr[j] > arr[j+1]){
                           int temp = arr[j];
                           arr[j] = arr[j+1];
                           arr[j+1] = temp;
                           swapped++;
                       }
                   }
                   if(swapped > 0){
                       break;
                   }
               }
}
```

B- O(n)
A- O(n²)
W- "

```
public static void printArr(int arr[]){
    for(int i=0; i< arr.length; i++){
        System.out.print(arr[i] + " ");
    }
    System.out.println();
}

public static void main(String args[]){
    int arr[] = {5,4,1,3,2};
    bubbleSort(arr);
    printArr(arr);
}
```

Selection Sort

↳ pick the smallest (from unsorted), put it at the beginning.

```
public static void selectionSort(int arr[]){
    for(int i=0; i<arr.length-1; i++){
        int minPos = i;
        for(int j=i+1; j<arr.length; j++){
            if(arr[minPos] > arr[j]){
                minPos = j;
            }
        }
        int temp = arr[minPos];
        arr[minPos] = arr[i];
        arr[i] = temp;
    }
    public static void main(String args[]){
        int arr[] = {5,4,1,2,3};
        selectionSort(arr);
        printArr(arr);
    }
```

B = O(n²)
A =
W =

unsorted part | t pla

```
        selectionSort(arr);
        printArr(arr);
    }
```

Insertion Sort — pick an element (from unsorted part) & place in the right pos in sorted part.

```
public static void insertionSort(int arr[]){
    for(int i=1; i<arr.length; i++){
        int curr = arr[i];
        int prev = i-1;
        while(prev>=0 && arr[prev] > curr){
            arr[prev+1] = arr[prev];
            prev--;
        }
        arr[prev+1] = curr;
    }
}

main(){
    insertionSort(arr);
    printArr(arr);
}
```